

NELSPRUIT, South AfricaWMO: **682890**Lat: **25.503S**Lon: **30.912E**Elev: **2897**StdP: **13.22**Time Zone: **2.00 (E02)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.8	44.9	13.4	12.1	65.1	19.4	16.2	66.7	28.4	57.0	17.0	57.2	3.5	260	0.292

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.1	89.9	63.8	87.2	64.3	84.7	64.8	72.1	81.0	70.9	79.7	69.9	78.2	7.6	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.5	120.9	75.8	68.2	115.6	74.7	67.3	111.9	73.9	37.9	81.9	36.7	79.9	35.8	78.5	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	11.3	9.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB													
			WB	34.7	74.8	3.1	4.2	32.4	77.8	30.6	80.2	28.9	82.5	26.6	85.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.6	71.7	72.4	71.4	67.1	63.7	59.8	58.8	62.1	66.0	67.2	69.0	70.3
	DBStd	6.51	3.89	3.34	4.02	4.34	4.65	4.47	4.09	5.17	6.28	5.83	5.43	4.56
	HDD50	2	0	0	0	0	0	2	0	0	0	0	0	0
	HDD65	738	4	0	6	28	79	162	199	120	65	42	23	10
	CDD50	6058	673	626	665	512	425	297	272	376	481	532	571	628
	CDD65	1319	212	207	206	90	39	7	5	31	96	109	144	173
	CDH74	8583	1066	1044	1092	480	308	116	90	410	1044	1004	1012	917
	CDH80	2191	223	212	241	89	44	12	8	93	376	359	328	206
(14) Wind	WSAvg	4.1	3.6	3.9	3.8	3.4	3.9	4.1	4.3	4.7	4.6	4.4	4.4	3.9
(15) Precipitation	PrecAvg	35.8	5.9	5.3	4.7	2.0	0.7	0.3	0.3	0.5	1.0	3.3	5.4	6.3
	PrecMax	54.4	12.3	15.8	7.6	3.5	1.7	1.6	1.1	1.5	3.6	6.8	9.4	12.7
	PrecMin	15.5	1.2	1.0	0.6	0.0	0.0	0.0	0.0	0.1	0.1	1.1	2.8	2.8
	PrecStd	8.6	3.0	3.8	2.0	0.9	0.5	0.4	0.3	0.4	0.9	1.2	1.8	2.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.4	88.8	89.2	86.8	84.3	82.2	80.8	86.8	92.5	94.4	92.3	89.2
		MCWB	71.6	68.0	66.5	63.3	59.1	53.9	53.4	56.6	59.7	62.0	63.5	69.3
	2%	DB	85.4	84.8	85.5	82.5	80.5	77.7	76.6	82.7	88.2	88.3	87.9	84.7
		MCWB	69.4	68.6	67.2	63.9	58.2	53.7	53.0	56.3	58.9	62.6	65.0	68.8
	5%	DB	82.5	82.6	82.6	79.2	77.3	73.8	73.2	78.9	84.5	83.9	83.5	81.9
		MCWB	68.8	67.9	67.3	64.5	57.9	53.3	52.9	55.8	58.8	62.9	66.6	68.3
	10%	DB	79.9	80.3	80.1	76.5	74.2	70.9	69.8	75.0	80.5	79.6	79.5	79.1
		MCWB	67.9	67.5	66.7	64.0	58.0	53.1	52.8	55.1	58.5	62.4	65.9	67.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.9	72.7	72.0	69.8	64.8	59.1	58.5	62.0	64.8	70.1	77.6	72.9
		MCDB	83.0	82.7	82.0	78.3	75.2	71.3	70.0	76.4	79.8	82.5	77.6	82.3
	2%	WB	71.7	71.1	70.4	68.1	62.7	57.5	56.9	60.4	63.1	66.6	70.5	71.0
		MCDB	81.2	80.3	79.7	75.5	72.5	68.9	67.7	73.8	77.0	79.5	78.3	79.8
	5%	WB	70.4	70.0	69.2	66.5	61.2	56.3	55.7	58.8	61.8	65.2	68.5	69.8
		MCDB	78.6	78.4	77.8	73.9	70.8	67.1	66.0	71.0	75.3	77.8	77.3	78.6
	10%	WB	69.3	69.0	68.2	65.3	59.9	55.2	54.6	57.3	60.5	63.8	67.1	68.6
		MCDB	76.5	76.7	76.2	73.0	69.3	65.6	64.9	68.9	73.8	75.1	75.5	76.7
(35) Mean Daily Temperature Range	5% DB	MDBR	14.2	15.1	15.8	16.6	20.7	21.8	21.2	21.7	21.5	18.2	16.1	14.5
		MCDBR	18.6	18.8	20.5	22.1	25.0	26.5	27.3	29.7	30.0	27.9	24.4	19.6
	5% WB	MCWBR	7.3	6.7	7.2	8.7	9.5	10.5	10.3	10.4	9.9	9.2	9.2	7.7
		MCWBR	15.7	16.0	16.7	17.8	19.7	20.6	20.1	22.0	22.4	21.4	17.9	16.9
(40) Clear-Sky Solar Irradiance	taub		0.413	0.403	0.398	0.377	0.338	0.321	0.323	0.387	0.441	0.468	0.430	0.431
		taud	2.333	2.360	2.363	2.382	2.449	2.472	2.455	2.270	2.149	2.126	2.264	2.274
	Ebn at Noon		296	293	285	277	275	274	278	269	268	272	289	291
		Edh at Noon	43	41	39	35	30	28	29	39	48	51	46	46
(44) All-Sky Solar Radiation	RadAvg		1912	1892	1677	1435	1332	1210	1289	1491	1688	1705	1813	1894
	RadStd		150	172	140	105	65	74	43	87	104	120	156	138

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page